

Nanoforge Product Portfolio

Power semiconductors and process-control solutions, engineered in Hyderabad and Singapore.

Nanoforge Semiconductor helps manufacturers across India and South-East Asia build power electronics and semiconductor capacity faster. We supply automotive-grade power devices and battery-management ICs to OEMs, and inline metrology, inspection and yield analytics to fabs and OSAT plants. Because we work on both sides of the chain - from the wafer to the inverter - our engineers understand what it takes to move from first silicon to stable volume production.

Power semiconductors

ForgeSiC 650 V / 1200 V MOSFET modules

Half-bridge and six-pack silicon carbide modules rated 200 A to 800 A. Silver-sintered die attach, low-inductance layout and AQG-324 qualification make them suitable for traction inverters in two-wheelers, three-wheelers and buses, as well as on-board chargers and 350 kW string inverters. Every module ships with die-level traceability, assembled and tested at qualified partner OSATs in India.

NF-GD gate drivers and power-management ICs

AEC-Q100 Grade 1 isolated gate drivers with desaturation detection, active Miller clamp and soft turn-off, paired with automotive buck/boost controllers and LDOs. Designed to drive ForgeSiC modules and third-party IGBTs with minimal external components.

CellGuard battery-management ICs

16-cell and 18-cell battery monitors with integrated passive balancing, isolated daisy-chain communication and ISO 26262 ASIL-C support. CellGuard scales from 48 V scooter packs to 800 V bus packs and multi-rack energy storage systems.

Reference designs

- 12 kW and 25 kW two-wheeler traction inverter
- 6.6 kW and 22 kW on-board charger
- 16S to 24S BMS board with state-of-charge firmware
- 3 kW GaN/SiC rack power shelf module for 48 V data-center racks

Fab and OSAT process control

InlineEye metrology and inspection

Film-thickness, overlay and critical-dimension metrology plus brightfield and darkfield defect inspection for 150 mm to 300 mm wafers, including silicon carbide substrates and epitaxial layers. Tools support SECS/GEM automation and are serviced from our Hyderabad and Penang depots with 24-hour on-site response.

Forgesight yield analytics

Forgesight combines inline metrology, inspection, parametric test and final test data into one model of your line. It flags excursions within minutes, ranks likely root causes and tracks yield learning curves across tools and lots. Deploy it on-premise or in a sovereign cloud; it connects to Nanoforge tools out of the box and to third-party tools through standard interfaces.

Why customers choose Nanoforge

- Local engineering. Field application engineers in Hyderabad, Bengaluru, Pune, Chennai, Singapore and Penang, plus a failure-analysis lab in Hyderabad.
- Allocation you can plan around. Multi-year allocation programmes with buffer stock held in India and Singapore.
- Localization-ready. Documentation to support domestic value-addition audits and trusted-source requirements.
- Quality first. IATF 16949 manufacturing, AEC-Q100/Q101 qualification and a 10-year supply commitment on automotive part numbers.

Who we serve

EV, two-wheeler and commercial vehicle OEMs; battery pack integrators; solar and storage inverter makers; telecom and data-center hardware companies; EMS providers; and new fabs, compound-semiconductor fabs and OSAT/ATMP plants.

Get started

Request samples, reference designs or a line assessment at nanoforge-semiconductor.example. Engineering samples and evaluation boards are available to qualified design projects, and our process-control team offers a free two-week yield baseline study for new fab and OSAT lines.